ST. LUKE'S HEALTH SYSTEM × DATABRICKS · ART OF THE POSSIBLE

Finding undiagnosed CKD, before it's missed

Chronic kidney disease is often **undocumented or mis-staged in Epic** — finding it today means manual chart review across thousands of patients. This surfaces those patients automatically and closes the gap, with a **clinician always in the loop**.

1

Governed data

Outpatient **labs**, medications, clinical notes & the Epic problem list — unified and governed in Unity Catalog.

→

2

Identify & suggest

KDIGO stage from lab trends, an **ML model** that suggests the stage + confidence, and AI that reads the notes for nephrology context.

→

3

Clinician worklist

A ranked review list in a Databricks App. The clinician opens a patient, sees the evidence, and **confirms or overrides** the stage.

→

4

Write back to Epic

Approval lands in a **governed, human-in-the-loop queue** — drained to Epic's problem list via FHIR / HL7 in production.

2,000	1,524	518	238	~0.98
Patients screened	Have CKD (by labs)	Undocumented in Epic	High-risk, not seeing nephrology	Avg model confidence

BUILT END-TO-END ON DATABRICKS

Unity Catalog governance + PHI masking · **Lakeflow** bronze → silver → gold · **AI Functions** read clinical notes · **MLflow + Model Serving** CKD stage model · **Metric Views** governed KPIs · **Genie** plain-English Q&A · **AI/BI Dashboards** · **Databricks Apps** clinician worklist · **Lakebase** Epic write-back queue

Human-in-the-loop — nothing auto-writes to Epic; a clinician approves every change.

Synthetic data · art of the possible — not yet connected to Epic or validated for clinical use.